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ABSTRACT

Graphene van der Waals (vdW) heterostructures, particularly those combined with hexagonal boron nitride (h-BN), exhibit unique electron–phonon interaction (EPI), enabling remarkable electron transport phenomena such as ultrahigh mobility, electron hydrodynamic flow, and superconductivity. Despite extensive studies on electron transport, the effect of EPI on phonon thermal transport in such heterostructures remains underexplored. In this Letter, we study the EPI-driven modulation of phonon thermal conductivity (k_{ph}) in the bilayer graphene/h-BN heterostructure via first-principles calculations. We find that k_{ph} varies nonmonotonically with carrier concentration due to the evolution of the Fermi surface near the Dirac point. The maximum reduction in k_{ph} compared to its intrinsic value reaches 41% at 300 K and 51% at 200 K, significantly exceeding the reduction reported for pristine graphene at a comparable carrier concentration. This significant reduction originates from the broken out-of-plane symmetry in the graphene/h-BN heterostructure, which enables direct flexural (ZA) phonon–electron coupling, and the strong EPI of in-plane shear (TA′) mode induced by the interlayer vdW interaction. A phonon branch-resolved analysis further shows that the relative contribution of ZA phonon–electrons scattering to the reduction in k_{ph} decreases from 68% to 25% with increasing carrier concentration, while the contributions from TA and TA′ phonon–electron scattering initially rise and eventually stabilize at around 30%. Our results provide insight into how EPI affects the thermal transport of graphene vdW heterostructures and offer guidance for thermal management in graphene-based nanodevices.

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Two-dimensional (2D) semiconductors are promising candidates for next-generation nanoelectronics,[1,2] enabling further miniaturization of field-effect transistors. As a pioneering 2D material, graphene has garnered tremendous attention due to its excellent electronic[3–5] and thermal[6–9] transport properties. However, its lack of a bandgap limits practical applications. This limitation can be improved by supporting/encapsulating graphene on a substrate.[10] Hexagonal boron nitride (h-BN), a 2D insulator, is an attractive supporting/encapsulating material for graphene, benefiting from its atomically flat surface[11,12] and a small lattice mismatch (∼2%) with graphene.[13] The resulting graphene/h-BN van der Waals (vdW) heterostructures have been extensively studied for their electronic transport properties,[14–19] and recent advances in synthesis techniques have facilitated their integration into high-performance devices.[11,14,20,21]

Given the critical role of thermal management in nanodevices, where extreme power densities and localized hotspots limit performance, reliability, and scalability, phonon thermal transport in graphene/h-BN heterostructures has also drawn significant attention. The interfacial thermal conductance (ITC) of bilayer graphene/h-BN heterostructure has been theoretically estimated to range from 32.5 to 50.0 MW m^{-2} K^{-1}, depending on the stacking configuration,[22] while the experimental value is as low as 7.4 MW m^{-2} K^{-1}, measured using the Raman spectroscopy technique.[23] Using molecular dynamics simulations, Ren *et al.* found that the low-frequency phonon transmission decreases with increasing interlayer rotation angle, resulting in a monotonic reduction in the ITC of graphene/h-BN.[24] Moreover, the interlayer coupling strength has been found to affect the interaction between out-of-plane and in-plane phonons in graphene/h-BN

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heterostructures and thus changes the ITC.[25] Considering the large contact area-to-thickness ratio and the high ITC in graphene/h-BN heterostructures, in-plane phonon thermal transport, where energy flows parallel to the vdW interface, is also critical for practical thermal management. However, research in this aspect is limited.

The electron–phonon interaction (EPI) is conventionally considered to be weak compared to phonon–phonon (ph–ph) scattering and is often neglected in the thermal transport analysis within semiconductors at room temperature and above. However, recent theoretical and experimental studies have demonstrated that EPI can significantly influence phonon thermal transport in materials with high electron concentration even at room temperature.[26–29] For example, Liao *et al.* found that the phonon thermal conductivity (k_{ph}) can be reduced by up to 45% in *p*-type silicon at a high carrier concentration of 10^{21} cm^{-3} via first-principles calculations.[26] Experimental evidence for the pronounced suppression of phonon propagation[29] and a significant reduction in k_{ph} on a nanosecond timescale[30] by EPI in silicon films at room temperature were subsequently observed using ultrafast photoacoustic spectroscopy and the transient thermal gating method, respectively. Group V transition metal carbides (TMCs) were demonstrated to have nearly temperature-independent k_{ph} due to anomalously strong intrinsic EPI relative to ph–ph interactions. This strong intrinsic EPI stems from the TMCs' nested Fermi surfaces and the large frequency gap between acoustic and optical phonons.[27] Yue *et al.* applied first-principles calculations to systematically investigate the phonon–electron (ph–el) scattering rates in 2D materials. They found that the ph–el scattering in 2D materials differs markedly from that in three-dimensional bulk materials due to reduced dimensionality and the resulting distinct electron and phonon dispersions.[28] Although direct scattering of ZA phonons with electrons is prohibited in monolayer graphene due to the out-of-plane reflection symmetry.[31] Yang *et al.* found that ZA phonons can indirectly interact with electrons via anharmonic coupling with TA and LA phonons.[32] This indirect interaction leads to a reduction in k_{ph} of up to 21% at room temperature. Moreover, the tunable electron–ZA phonon interaction depending on temperature was experimentally observed in the h-BN/graphene/h-BN heterostructures.[16] Despite extensive studies on the influence of EPI on phonon thermal transport in both bulk and individual 2D materials, its effect on 2D vdW heterostructures has yet to be studied.

In this Letter, we investigate the effect of EPI on k_{ph} of the bilayer graphene/h-BN heterostructure via first-principles calculations. Among various stacking configurations, the AB stacking, where one carbon (C) atom sits above a boron (B) atom and the other C atom is centered above the h-BN hexagon, is the most stable configuration.[10,33,34] Therefore, we focus on this stacking arrangement in this study, as shown in Fig. 1(a).

Density functional theory (DFT) calculations are performed using the Quantum ESPRESSO package.[35] The scalar-relativistic optimized norm-conserving Vanderbilt (ONCV)[36] pseudopotential within the local density approximation (LDA) is employed.[37] The chosen exchange-correlation approach is widely used for studies on the effect of EPI on k_{ph}.[32,38,39] The vdW correction (rVV10)[40] for the interlayer interaction between graphene and h-BN is also considered. To avoid the artificial interaction, a vacuum layer with a total thickness of 20 Å is added along the *c*-axis. The plane wave kinetic energy cutoff and the charge density cutoff are set to 90 and 360 Ry, respectively. The total electron energy convergence threshold for self-consistency is set to 1×10^{-11} Ry. In the ground-state calculations, the Brillouin zone is sampled with a mesh size of $56 \times 56 \times 1$. The lattice constant of the graphene/h-BN heterostructure is 2.485 Å, consistent with values reported in previous studies.[33,41,42] The interlayer distance *d* is 3.35 Å, also in good agreement with both experimental and theoretical results.[15,41,43] The second-order interatomic force constants (IFCs) are calculated using density functional perturbation theory (DFPT) on an $8 \times 8 \times 1$ $\boldsymbol{q}$ grid. The third-order IFCs are obtained via the finite displacement method using a $10 \times 10 \times 1$ supercell, including interactions up to the 9th nearest neighbor. The electron and phonon Boltzmann transport equations (BTEs) are iteratively solved via an open-source code Elphbolt,[44] which is widely used for studies of coupled phonon–electron transport.[39,45,46] The detailed solution process

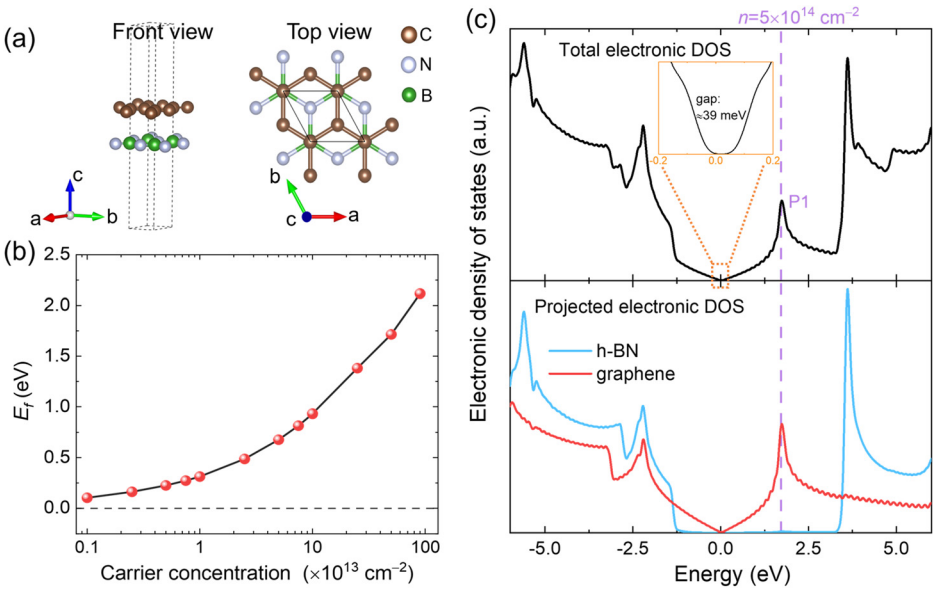


FIG. 1. (a) Schematic illustration of the graphene/h-BN van der Waals heterostructure. (b) Carrier concentration as a function of Fermi energy (E_f). (c) Electronic density of states (DOS) of the graphene/h-BN heterostructure. The upper panel is the total DOS, and the lower panel presents the projected DOSs of graphene (red) and h-BN (blue). A zoomed-in inset reveals the opening of a small gap of approximately 39 meV.

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and method can be found elsewhere.[44,47] The electron–phonon (el–ph) matrix elements are computed using the EPW package,[48] employing a coarse $56 \times 56 \times 1$ $\boldsymbol{k}$ mesh and an $8 \times 8 \times 1$ $\boldsymbol{q}$ grid. These el–ph matrix elements, expressed in Wannier space, are then used as an input for transport calculations in Elphbolt, where they are transformed into Bloch representation on a fine $300 \times 300 \times 1$ $\boldsymbol{k}$ mesh and a $100 \times 100 \times 1$ $\boldsymbol{q}$ grid. The phonon and electron dispersions, along with the convergence test of the calculated k_{ph} are provided in the supplementary material. The four-phonon scattering is neglected, as its impact on k_{ph} of graphene/h-BN is expected to be negligible at or below room temperature due to broken reflection symmetry and the splitting of the ZA mode (supplementary material), as evidenced in bilayer graphene[49] and bulk h-BN.[50]

The total scattering rate ($1/\tau_{b\boldsymbol{q}}^{total}$, the inverse relaxation time) of each phonon mode (b, $\boldsymbol{q}$), where b and $\boldsymbol{q}$ represent phonon branch index and wavevector, respectively, is calculated by combining phonon–electron (ph–el) ($1/\tau_{b\boldsymbol{q}}^{ph-el}$) and ph–ph ($1/\tau_{b\boldsymbol{q}}^{ph-ph}$) scattering rates using Matthiessen's rule,[51]

$$\frac{1}{\tau_{b\boldsymbol{q}}^{total}} = \frac{1}{\tau_{b\boldsymbol{q}}^{ph-el}} + \frac{1}{\tau_{b\boldsymbol{q}}^{ph-ph}}, \tag{1}$$

and $\tau_{b\boldsymbol{q}}^{ph-el}$ is calculated as

$$\frac{1}{\tau_{b\boldsymbol{q}}^{ph-el}} = \frac{2\pi}{\hbar}\frac{2}{1+n_{b\boldsymbol{q}}^{0}}\sum_{mn\boldsymbol{k}}\left|G_{\boldsymbol{kq}}^{bmn}\right|^{2} f_{m\boldsymbol{k}}^{0}\left(1-f_{n\boldsymbol{k}+\boldsymbol{q}}^{0}\right)\delta \times \left(\varepsilon_{n\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{m\boldsymbol{k}} - \hbar\omega_{b\boldsymbol{q}}\right), \tag{2}$$

where f^0 is the Fermi–Dirac equilibrium distribution of electrons; n^0 is the Bose–Einstein equilibrium distribution of phonons; ε is the electron energy; $\hbar$ and ω represent the reduced Planck's constant and phonon angular frequency, respectively; δ indicates the Dirac delta function. $\left|G_{\boldsymbol{kq}}^{bmn}\right|$ is the el–ph matrix element, depicting that an electron in the initial state $m\boldsymbol{k}$ is scattered by a phonon with mode (b, $\boldsymbol{q}$) to the final state $n\boldsymbol{k}+\boldsymbol{q}$. k_{ph} affected by the ph–el scattering, is then given by

$$k_{ph} = \frac{1}{N_{\boldsymbol{q}}\Omega K_B T^2}\sum_{b\boldsymbol{q}} n_{b\boldsymbol{q}}^{0}\left(n_{b\boldsymbol{q}}^{0}+1\right)\left(\hbar\omega_{b\boldsymbol{q}}\right)^{2}\boldsymbol{v}_{b\boldsymbol{q}}\otimes\boldsymbol{F}_{b\boldsymbol{q}}, \tag{3}$$

where $N_{\boldsymbol{q}}$ and Ω are the number of sampled $\boldsymbol{q}$ points and the unit cell volume, respectively; K_B represents the Boltzmann constant; $\boldsymbol{v}$ is the phonon group velocity. The phonon effective mean free path $\boldsymbol{F}$ is determined by the linearized phonon BTE,[32,52]

$$\boldsymbol{F}_{b\boldsymbol{q}} = \left(\boldsymbol{v}_{b\boldsymbol{q}} + \Delta_{b\boldsymbol{q}}\right)\tau_{b\boldsymbol{q}}^{total}, \tag{4}$$

where $\Delta_{b\boldsymbol{q}}$ denotes the effective variation in phonon velocity induced by phonon scattering events.[52]

For practical applications, carrier concentration in 2D semiconductors can be modulated via an electrostatic gate.[28,53,54] In our calculations, the carrier (electron) concentration is controlled by tuning the Fermi energy (E_f). The relationship between carrier concentration and E_f in the graphene/h-BN heterostructure is shown in Fig. 1(b). The highest carrier concentration considered in this work is $9 \times 10^{14}\,\mathrm{cm}^{-2}$, which is expected to be experimentally within reach using an ionic gate technique,[53,55] where a carrier concentration as high as $4 \times 10^{14}\,\mathrm{cm}^{-2}$ has already been achieved.[53] The total and projected electronic density of states (DOS) are shown in Fig. 1(c), where the projected DOS distinguishes contributions from graphene (red) and h-BN (blue). As shown in the zoomed-in inset of Fig. 1(c), the bandgap in graphene/h-BN is around 39 meV, consistent with that reported in previous studies.[10,16] Since h-BN has a wide bandgap of approximately 5 eV, the maximum E_f of ≈2.1 eV investigated in this study corresponds to a carrier concentration of $9 \times 10^{14}\,\mathrm{cm}^{-2}$ and remains well below the conduction band edge of h-BN. Therefore, h-BN is not expected to conduct electrons at the maximum carrier concentration in the calculations.

As shown in Fig. 2(a), k_{ph} of the graphene/h-BN heterostructure is influenced by EPI and exhibits a nonmonotonic dependence on carrier concentration. When $n < 1 \times 10^{13}\,\mathrm{cm}^{-2}$, k_{ph} is weakly affected by EPI and remains close to the intrinsic value. However, when n exceeds $1 \times 10^{14}\,\mathrm{cm}^{-2}$, k_{ph} decreases sharply, reaching a minimum at $n = 5 \times 10^{14}\,\mathrm{cm}^{-2}$. Beyond this point, k_{ph} begins to increase with n. Because phonons have a much smaller energy scale compared to electrons, the energy difference between the initial and final electron states involved in a scattering process is nearly negligible [see the Dirac delta function in Eq. (2), which enforces both energy and momentum conservation for ph–el scattering]. Therefore, ph–el scattering events can be assumed as an "on-shell" process.[28] Consequently, the nonmonotonic behavior of k_{ph} is related to the sizes and shapes of Fermi surfaces at different carrier concentrations, as reflected in the electronic DOS of graphene/h-BN heterostructure [Fig. 1(c)]. When $n < 5 \times 10^{14}\,\mathrm{cm}^{-2}$, corresponding to the left-side P1 peak labeled in electronic DOS in Fig. 1(c), the electronic DOS in the conduction band increases with n, which means the larger Fermi surface, enabling more low-frequency phonons to participate in ph–el scattering events (also evidenced by Fig. 4). Moreover, the closer E_f is to the P1 peak, the steeper the slope of the electronic DOS curve becomes. As a result, k_{ph} declines more rapidly as n increases. As observed in Figs. 1(c) and 2(a), within the range of n considered in this work, the electronic DOS reaches its maximum at $n = 5 \times 10^{14}\,\mathrm{cm}^{-2}$, corresponding to the strongest EPI, and, thus, k_{ph} has its minimum value. Once n exceeds $5 \times 10^{14}\,\mathrm{cm}^{-2}$, the electronic DOS begins to decrease, weakening ph–el scattering as n increases, which leads to an increase in k_{ph}.

Given the not well-established definition for effective thickness of 2D materials, which is defined as twice the interlayer distance of the graphene/h-BN in this work, we further estimate the reduction percentage in k_{ph} caused by EPI, compared to the intrinsic k_{ph} of the graphene/h-BN heterostructure. As shown in Fig. 2(b), the effect of EPI on k_{ph} decreases with increasing temperature. This trend is expected because the ph–ph scattering becomes more prominent and dominant at higher temperatures, while EPI exhibits negligible temperature dependence.[32,56] As a result, the relative impact of EPI on k_{ph} diminishes with increasing temperature. Notably, k_{ph} is reduced by up to 41% at $n = 5 \times 10^{14}\,\mathrm{cm}^{-2}$ compared to the intrinsic value at 300 K, which is significantly greater than the 21% reduction observed in monolayer graphene at an approximately equal carrier concentration.[32] Additionally, the small change in interlayer distance does not affect the nonmonotonic k_{ph} induced by EPI (see the supplementary material). We attribute this enlarged reduction in k_{ph} to the broken reflection symmetry in the graphene/h-BN heterostructure, which relaxes the restrictive selection rule and enables direct interaction between ZA phonons and electrons[16] [see Fig. 4(a)]. Moreover, the polar B–N bond can induce electronic polarization within the

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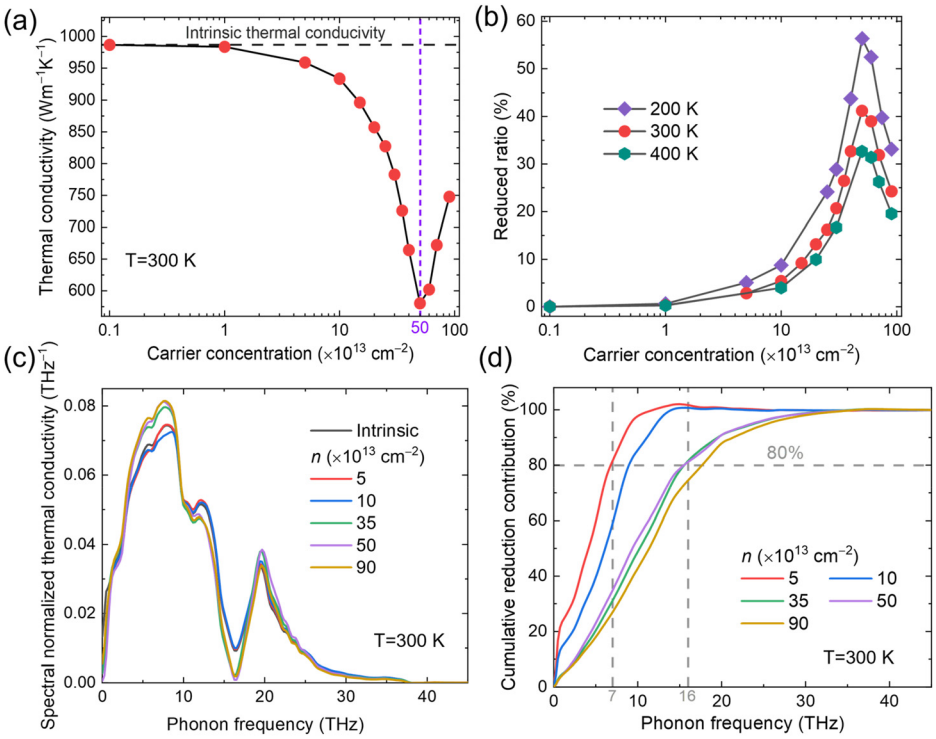


FIG. 2. (a) k_{ph} as a function of carrier concentration at room temperature. The intrinsic k_{ph} is also indicated by the black horizontal dashed line. (b) Relative reduced ratio of k_{ph} at various carrier concentrations compared to the intrinsic value. (c) Spectral normalized thermal conductivity [$S_{normalized_k}(w)$] for different carrier concentrations at 300 K. (d) Cumulative reduction contribution.

graphene/h-BN heterostructure (see the supplementary material), which may also lead to the stronger EPI.[34]

To identify the effect of phonon with different frequencies on the thermal conductivity, we define spectral normalized thermal conductivity ($S_{normalized\ k}$) as

$$S_{normalized_k}(w) = \frac{d\left(\frac{k_{c,w}}{k_{ph}}\right)}{d(w)}, \tag{5}$$

where $k_{c,w}$ represents the cumulative thermal conductivity (supplementary material) up to phonon frequency w. k_{ph} is the phonon thermal conductivity of the graphene/h-BN heterostructure. As shown in Fig. 2(c), k_{ph} of the graphene/h-BN heterostructure is mainly dominated by phonons below 15 THz. Notably, phonons near 7 THz contribute more significantly, which may be attributed to that these phonons have large populations due to thermal excitations at 300 K (corresponding to a thermal energy of k_BT). Moreover, compared to low carrier concentrations ($n < 1 \times 10^{14}$ cm^{-2}), the contribution of medium-frequency (10–16 THz) phonons to k_{ph} decreases at higher carrier concentrations, whereas the contribution of low-frequency (<7 THz) phonons increases. This trend results from the phonon frequency-dependent ph–el scattering. As shown in Fig. 4, ph–el scattering rates for low-frequency phonons remain largely unchanged across carrier densities. However, for medium-frequency phonons (10–16 THz), the ph-el scattering rates at high carrier densities are 5–6 orders of magnitude higher than at low densities, thereby suppressing thermal conductivity. Consequently, the contribution of medium-frequency phonons to k_{ph} decreases with higher carrier densities, while the contribution from low-frequency phonons increases. We further calculate the cumulative reduction contribution, defined as $(k_{c,i} - k_{c,n})/(k_{ph,i} - k_{ph,n})$, where c and i represent cumulative and intrinsic, respectively. As shown in Fig. 2(d), at low carrier concentrations ($n < 1 \times 10^{14}$ cm^{-2}), EPI of low-frequency (<7 THz) phonons is primarily responsible for the reduction in k_{ph}, contributing ≈80% to this effect. However, as the carrier concentration increases, more high-frequency phonons can participate in scattering events, with electrons and phonons in the range of 10–16 THz contributing over 30% to the reduction when $n > 3.5 \times 10^{14}$ cm^{-2}.

The branch-resolved thermal conductivities are further analyzed as a function of carrier concentration. Since phonons above 30 THz have an ignorable contribution to k_{ph} [Fig. 2(c)], we focus on the ZA, LA, TA, LA′, TA′, and ZO′ phonon branches (see the phonon dispersion in the supplementary material). Here, LA′/TA′ corresponds to the shear mode, while ZO′ represents the breathing mode.[57,58] These phonon modes are visualized at the Γ point in Fig. 3(a), combining an open-source code[59] with VESTA.[60] As shown in Fig. 3(b), EPI has a more profound impact on the thermal conductivity of TA, ZA, and TA′ branches, while it has a trivial influence on that of LA and LA′. The percentage contribution of different phonon branches to the total k_{ph} of the graphene/h-BN heterostructure is further calculated in Fig. 3(c). Unlike pristine graphene, where ZA phonons dominate thermal transport,[61] in the graphene/h-BN heterostructure, the TA branch contributes most to k_{ph}. Additionally, when $n > 2 \times 10^{14}$ cm^{-2}, the percentage contribution from the TA′ branch decreases significantly as n increases, indicating that at high carrier concentrations, the TA′ phonon transport is more significantly suppressed by EPI compared to other phonon branches. Figure 3(d) shows the percentage contribution

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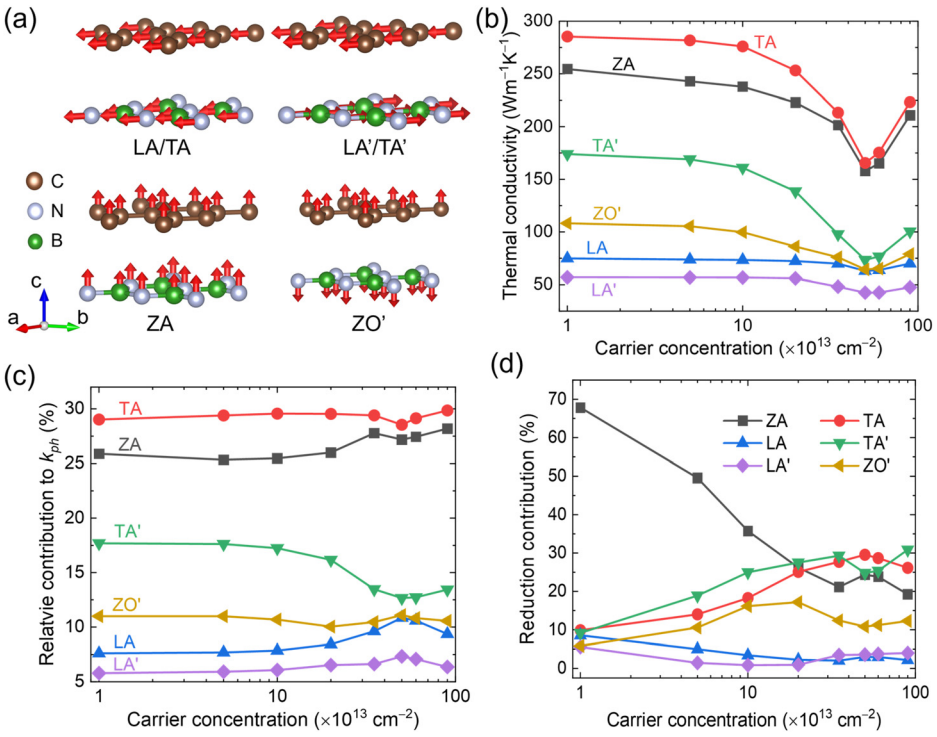


FIG. 3. (a) The atomic vibration patterns of C (brown), N (gray), and B (green) atoms at the Γ point for the LA/TA, LA′/TA′, ZA, and ZO′ modes. (b) Branch-resolved thermal conductivity as a function of carrier concentration. (c) Branch-resolved contribution to k_{ph}. (d) Branch-resolved reduction contribution to the total reduction in k_{ph}.

of each phonon branch to the reduction in k_{ph}, calculated as $(k_{i,branch} - k_{n,branch})/(k_{i,total} - k_{n,total})$, where i and n denote intrinsic and carrier concentration, respectively. Notably, the ph-el scatterings from ZA mode are the main contributors to the reduction in k_{ph} when $n \leq 1 \times 10^{14}\,\mathrm{cm}^{-2}$, which is consistent with the results in Fig. 2(d) where low-frequency phonons (<10 THz) account for ≈90% of the reduction in k_{ph} at the low carrier concentration. However, its contribution decreases rapidly from 68% as the carrier concentration increases, and eventually stabilizes at around 25% when $n > 2 \times 10^{14}\,\mathrm{cm}^{-2}$. As the carrier concentration increases from 1×10^{13} to $2 \times 10^{14}\,\mathrm{cm}^{-2}$, the contributions of the TA and TA′ modes rise from ≈10% to ≈30%.

To gain a deeper understanding of the physical mechanisms behind the reduction in k_{ph} of the graphene/h-BN heterostructure induced by EPI, we further calculate the ph–el and ph–ph scattering rates. Figure 4 reveals that ph–el scattering rates for all phonon branches exhibit a nonmonotonic dependence on carrier concentration, directly evidencing that the predicted nonmonotonic k_{ph} behavior originates from carrier-concentration-dependent EPI. Unlike in the pristine graphene, where electron–ZA phonon interaction is symmetry-forbidden, ZA phonons in the graphene/h-BN heterostructure can be scattered by electrons directly due to broken reflection symmetry, as shown in Fig. 4(a). Notably, for low-frequency ZA phonons (<1 THz), the ph–el scattering rate is comparable to the ph–ph scattering rate at a low carrier concentration ($n \leq 1 \times 10^{14}\,\mathrm{cm}^{-2}$). However, for the ZA phonon with higher frequencies, ph–ph scattering is significantly stronger than ph–el scattering, regardless of carrier concentrations. In contrast, for TA and TA′ modes, ph–el scattering rates become comparable to or even exceed ph–ph scattering rates at elevated carrier concentrations [Figs. 4(b) and 4(d)]. This explains the observed transition in reduction contribution in phonon branches: ZA mode relative contributions diminish with increasing carrier concentration, and TA/TA′ mode contributions become dominant [Fig. 3(d)]. For LA and LA′ modes [Figs. 4(c) and 4(e)], the ph–ph scattering rate is at least one order of magnitude higher than ph–el scattering rates. As a result, the total scattering rate remains basically unchanged based on Eq. (1), and the branch-resolved thermal conductivities of these modes are nearly EPI-independent. Although the ph–el scattering rates of the ZO′ mode are comparable to, or even exceed, the ph-ph scattering rates at high carrier concentrations [Fig. 4(f)], its effects on k_{ph} remain small because the ZO′ mode contributes only about 10% to k_{ph} [Fig. 3(c)].

For metallic and semiconducting materials, the electronic contribution to thermal conductivity (k_e) must be taken into account, especially at high carrier concentrations.[32] Therefore, we also calculate the total thermal conductivity ($k_{total} = k_e + k_{ph}$) (see the supplementary material), where a maximum reduction of 34% in k_{total} is still observed at 300 K. Furthermore, the external electric field-induced screening, which is not captured in this work, may further reduce k_{ph} in graphene/h-BN heterostructures due to phonon renormalization, as supported by first-principles calculations in bilayer graphene, monolayer silicene, and germanene.[62,63] Moreover, we note that h-BN/graphene/h-BN configurations are also common in graphene-based devices. The effect of EPI on k_{ph} is expected to be still significant, considering the existing electron-ZA coupling[16] and enhanced EPI due to flat bands on the Brillouin zone edge of the reconstructed graphene electronic band.[64,65] The twist angle not only tunes the electron–optical phonon interaction,[65,66] but also renormalizes phonon dispersion. Consequently, the effect of EPI on thermal transport in twisted graphene/h-BN heterostructures warrants further investigation, which is beyond the scope of this study.

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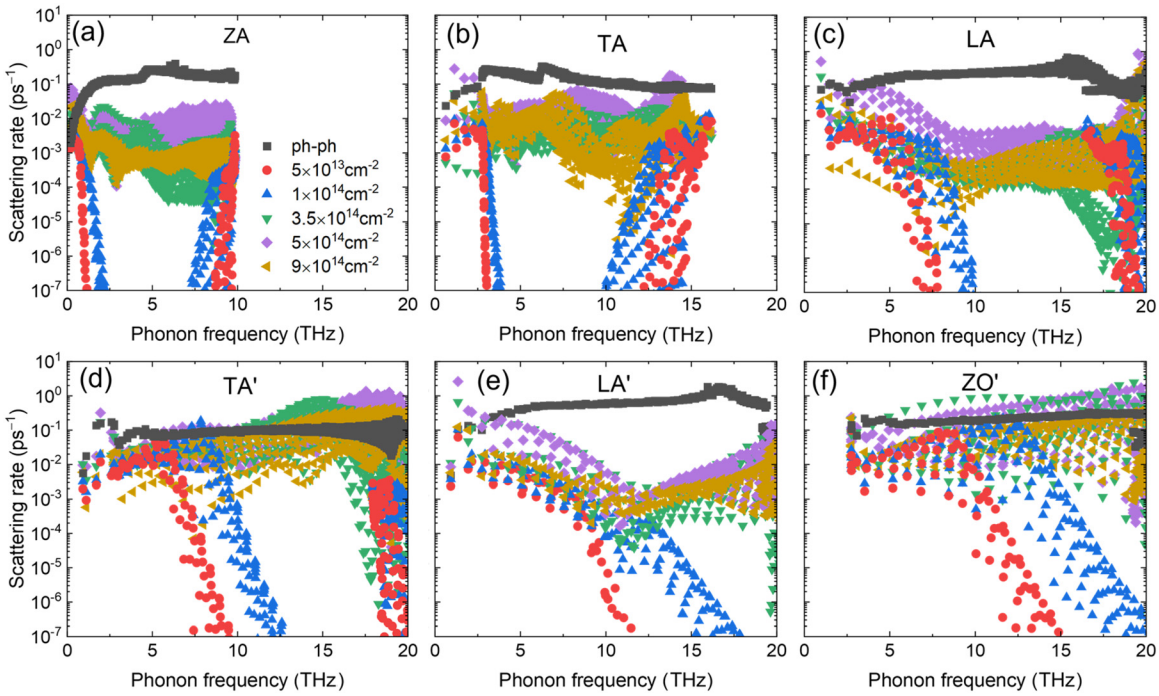


FIG. 4. Phonon–electron scattering rates of (a) ZA, (b) TA, (c) LA, (d) TA′, (e) LA′, and (f) ZO′ modes at different carrier concentrations. Phonon–phonon scattering rates (black rectangles) for each mode are also included.

In summary, we investigate the effects of EPI on phonon thermal transport in bilayer graphene/h-BN heterostructures using first-principles calculations. In the calculated carrier concentration range, k_{ph} initially decreases and then increases with rising carrier concentration due to Fermi surface evolution near the Dirac point. The broken out-of-plane reflection symmetry enables direct coupling between ZA phonons and electrons, accounting for the observed reduction in k_{ph} at low carrier concentrations. At high carrier concentrations, enhanced coupling of electrons with TA and TA′ phonons, owing to increased participation of high-frequency phonons in EPI, leads to their greater contribution to the reduction in k_p. The modified thermal transient gating method introduced by Ref. 30 has the potential to experimentally observe the large EPI effect on graphene/h-BN heterostructures. This study provides fundamental insights into EPI-mediated thermal transport modulation in 2D vdW heterostructures and offers valuable guidance for thermal management in 2D semiconductor devices.

See the supplementary material for more details on phonon and electron dispersions, convergence test for k_{ph}, the robustness testing of interlayer distance, the electronic differential charge-density distribution, the cumulative thermal conductivity for different carrier concentrations at 300 K, and the total thermal conductivity ($k_{total} = k_e + k_{ph}$).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Ziwen Zou: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead). **Hongqi Tao:** Investigation (supporting); Supervision (supporting). **Jingwen Zhang:** Writing – original draft (supporting). **Ruinan Wu:** Supervision (supporting). **Likang Cai:** Supervision (supporting). **Zhe Cheng:** Supervision (supporting). **Menglong Hao:** Conceptualization (equal); Funding acquisition (lead); Project administration (lead); Supervision (lead); Validation (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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